

6_Evaluación_resultados

February 24, 2026

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[1]: # Cargar los datos del test
import numpy as np
test = np.load('../final_features/test.npy')

X_test = test[:, :-1]
y_test = test[:, -1]

print(np.shape(X_test))
print(np.shape(y_test))
```

(37, 10)

(37,)

```
[4]: # Cargamos los modelos entrenados
import pickle
with open('../models/LOGR.pickle', 'rb') as fr:
    LOGR = pickle.load(fr)
with open('../models/MLP.pickle', 'rb') as fr:
    MLP = pickle.load(fr)

# Extraer las predicciones
y_pred_LOGR = LOGR.predict(X_test)
y_pred_MLP = MLP.predict(X_test)

y_pred = [y_pred_LOGR, y_pred_MLP]

# Evaluar diversas métricas de clasificación
from sklearn import metrics

from tabulate import tabulate
headers = ['', 'LOGR', 'MLP']
P, S, FS, ACC, AUC = [['Precision'], ['Sensibilidad'], ['F1-Score'],
↪ ['Accuracy'], ['AUC']]

for i in range(0,2):
    P.append(np.round(metrics.precision_score(y_test, y_pred[i]),4))
    S.append(np.round(metrics.recall_score(y_test, y_pred[i]),4))
    FS.append(np.round(metrics.f1_score(y_test, y_pred[i]), 4))
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    ACC.append(np.round(metrics.accuracy_score(y_test, y_pred[i]), 4))
    AUC.append(np.round(metrics.roc_auc_score(y_test, y_pred[i]), 4))

my_data = [tuple(P), tuple(S), tuple(FS), tuple(ACC), tuple(AUC)]
print(tabulate(my_data, headers=headers))

# Confusion matrix
print(metrics.confusion_matrix(y_test, y_pred[1]))

```

	LOGR	MLP
Precision	0.8421	0.8889
Sensibilidad	1	1
F1-Score	0.9143	0.9412
Accuracy	0.9189	0.9459
AUC	0.9286	0.9524

```

[[19  2]
 [ 0 16]]

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